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The Interaction between Credit Constraints and Uncertainty Shocks

This paper proposes a novel link between credit markets and uncertainty shocks. We introduce a role for credit uncertainty via collateral constraints in an otherwise standard real business cycle (RBC) model and show that an increase in credit uncertainty triggers a precautionary response that interacts with the collateral constraint to generate a simultaneous decline in output, consumption, investment, real wages, and hours; a feature that previous work on uncertainty shocks without credit constraints is unable to produce in a flexible-price environment. We also empirically test the theoretical predictions and show that an unforeseen increase in credit uncertainty generates a simultaneous decline in a broad measure of real activity in recessions.

JEL codes: C32, E32, E44

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The importance of uncertainty shocks in generating business cycle fluctuations has been a growing area of research since the Great Recession (e.g., Bloom 2009, Jurado, Ludvigson, and Ng 2015), with policymakers frequently attributing the slower recovery following the recession to elevated macroeconomic uncertainty.[1] Concurrently, the role of credit markets in shaping business cycle dynamics has gained traction, with many authors documenting a link between credit buildup in periods of expansion and the subsequent crash in recessions (e.g., Jordà, Schularick, and Taylor 2013, Mian and Sufi 2018). This paper studies the interaction of these two channels by proposing a novel link between credit markets and uncertainty shocks.

We establish time-varying uncertainty in credit growth rate as a robust empirical feature. Then, we use a parsimonious model to explain how shocks to credit uncertainty can generate business cycle comovement in macro-economic variables in a flexible-price dynamic stochastic general equilibrium (DSGE) environment. This feature of business cycle comovement is challenging to generate without other frictions, such as nominal rigidities. In a standard real business cycle (RBC) model, uncertainty shocks (introduced through time-varying volatility in shocks to aggregate demand or supply) are expansionary for hours, investment, and output but contractionary for consumption and wages. Our paper introduces a role for credit uncertainty via a collateral constraint channel in an otherwise standard RBC model. Here, when the collateral constraint binds, an increase in credit uncertainty is contractionary for hours, wages, consumption, investment, and output. This feature of business cycle comovement is an outcome of the interaction between credit uncertainty, precautionary motives, and the collateral constraint. The proposed channel thus generates a sizable and simultaneous decline in macro variables of interest in a flexible-price environment.

How do unforeseen changes in credit uncertainty transmit in the model to generate comovement in the response of macro variables? A shock to credit uncertainty in the model causes households to cut consumption and increase labor supply. The former response increases the marginal utility of consumption and precautionary savings. While the precautionary-driven increase in labor supply implies that the household is willing to work more at a given wage or accept lower wages at given hours. This household response is expansionary for hours and investment while contractionary for consumption.

On the firm side, the wedge introduced by the collateral constraint limits working capital financing by firms and distorts the equilibrium condition in the labor market. A shock to credit uncertainty increases this wedge, further distorting equilibrium condi-

1. Fisher (2013) and Lagarde (2015) explicitly mention elevated economic uncertainty leading to a slower recovery following the Great Recession. More recently, Gopinath (2020) addressed the role of uncertainty during the COVID-19 pandemic.

tions describing labor demand and depressing the price of capital. Consequently, the increase in the collateral-constraint-induced wedge generates a decline in labor demand. If the decline in labor demand is strong enough, it more than offsets the initial precautionary-induced expansion in labor supply; this causes a fall in hours and real wages in equilibrium. The fall in the price of capital generates a fall in investment, and given this is a closed-economy model, this causes a fall in output. This interaction between credit uncertainty, collateral constraints, and precautionary motives can thus successfully generate comovement in macro variables of interest.

Next, we test if, empirically, changes in credit uncertainty can generate a simultaneous decline in real variables that we find in the model. We identify shocks to credit uncertainty using the estimated shocks to the latent volatility underlying the credit growth rate in the United States. To quantify the real effects of credit uncertainty in the data, we use the method of local projections. We follow Jordà (2005) and use a linear model to understand the average effects of changes in credit uncertainty over the business cycle along with a state-dependent extension of local projections (similar to Ramey and Zubairy 2018) to examine the differential effects in recessions and expansions.

In the linear model, a one-standard-deviation increase in credit uncertainty generates a slowdown across a broad measure of real activity. In this way, our estimated shock propagates similarly to uncertainty shocks derived from other sources, for example, VIX, JLN measure of uncertainty in Jurado, Ludvigson, and Ng (2015). The novel feature of our empirical results lies in quantifying the importance of state dependence in examining the real effects of shocks to credit uncertainty. The linear model suggests that a shock to credit uncertainty is contractionary; however, once we allow for state dependence, we find that relative to the linear model, the effects are significantly larger in recessions and with almost no effects in expansions.

During downturns, a one-standard-deviation increase in credit uncertainty triggers a sharp and significant decline in auto sales, durable goods consumption, and investment. More broadly, aggregate consumption and GDP contracts; the unemployment rate shows a significant increase a few periods after the shock hits, and nondurable consumption declines. In expansions, the effects are mild[2] and often insignificant for several macro-economic variables.[3] Upswings in credit uncertainty in booms, therefore, *do not* impede the pace of real activity. During busts, however, an increase in credit uncertainty t can exacerbate the depth and duration of a recession. Therefore, we empirically establish this asymmetry in how credit uncertainty propagates in the economy.

We conclude our analysis by examining the state-dependent effects of uncertainty in the theoretical model to outline the impact of recessions and expansions in transmitting shocks to credit uncertainty. To study state dependence in the DSGE model, we

2. Slight contraction in aggregate consumption, nondurable goods consumption, GDP, and non-farm employment.

3. Auto sales, durable goods consumption, investment, and unemployment.

solve the fully nonlinear model using global methods and allow the credit constraint to bind occasionally. We then show that as the distance to slackness increases (capturing the transition from expansions to recessions), real variables record a sharper decline in response to increases in credit uncertainty in this fully nonlinear environment. This way, the model can generate the qualitative pattern and the asymmetric effects we find in the data.

Related literature: The first group of studies connected to our paper examines the role of uncertainty in generating business cycle fluctuations. Bloom (2009), Fernández-Villaverde et al. (2015), Leduc and Liu (2016), Basu and Bundick (2017), Berger, Dew-Becker, and Giglio (2020), Fernández-Villaverde and Guerrón-Quintana (2020), and Born and Pfeifer (2021) among others, motivate the role of shocks to the second moment as a driver of business cycles.[4]

One of the challenges in the theoretical literature examining the effects of uncertainty is generating a sizable impact and simultaneous decline in key macro variables—output, hours, wages, consumption, and investment. Our approach to modeling uncertainty and its interaction with the real economy thus overcomes the quantitative irrelevance result of time variation in uncertainty in RBC models described in Fernández-Villaverde and Guerrón-Quintana (2020) and can better explain the simultaneous decline in these variables than existing models of uncertainty shocks.

Our work is also related to empirical studies that examine the role of financial conditions in transmitting uncertainty shocks. Caldara et al. (2016), Alessandri and Mumtaz (2019), Ludvigson, Ma, and Ng (2021), and Caggiano et al. (2020) use sequential identification, regime differentiation, and shock-based restrictions to examine the role of financial markets in transmitting uncertainty. Instead of imposing identifying restrictions and regime dependencies, we construct our measure of uncertainty by exploiting the dynamics underlying credit expansion and contraction.

A third related strand of the literature comprising studies such as Jermann and Quadrini (2012), Liu, Wang, and Zha (2013), Christiano, Motto, and Rostagno (2014), Arellano, Bai, and Kehoe (2019), among others, examine the importance of financial factors during the Great Recession. Other studies, such as Alfaro, Bloom, and Lin (2024), consider the size of the finance uncertainty multiplier in the transmission of uncertainty. Our work provides a complementary view of the role of financial factors by isolating the role of changes in credit uncertainty.

The rest of the paper is organized as follows. Section 1 describes the data, models, and estimation methods used to uncover time variation in credit uncertainty. Section 2 provides intuition to interpret changes in the latent volatility underlying the credit growth rate estimated in Section 1. Section 3 introduces a parsimonious model that examines how changes in credit uncertainty can impact real activity and generate

4. Uncertainty in these papers stems from time-varying volatility in exogenous shocks to aggregate productivity, aggregate demand, fiscal policy, or from a realized volatility perspective. See Fernández-Villaverde and Guerrón-Quintana (2020) for a detailed review of the literature on uncertainty shocks and business cycle research.

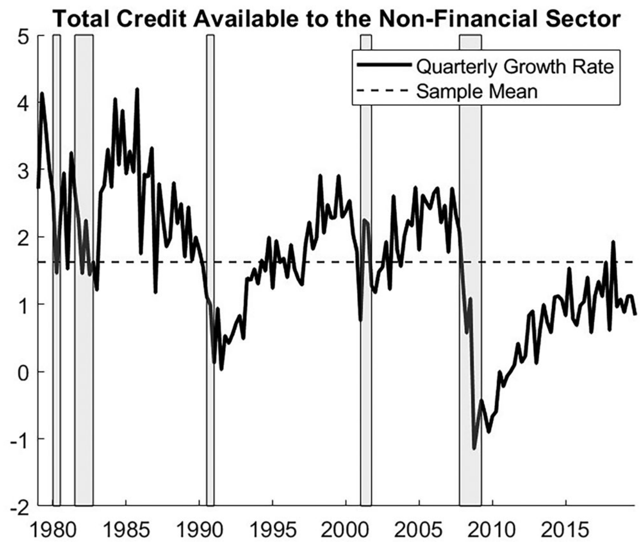

Fig 1. Total Credit Extended to the Nonfinancial Sector.

NOTES: Shaded areas indicate NBER recessions.

a simultaneous decline in output, investment, consumption, hours, and real wages in a flexible-price model. Section 4.1 tests the empirical validity of the theoretical predictions using the method of local projections. Section 4.2 examines state dependence in the theoretical model. Section 5 concludes. The paper has an online supplement containing further technical results and robustness checks to the theoretical and empirical findings we present in the main paper.

1. TESTING THE EXISTENCE OF STOCHASTIC VOLATILITY (SV)

1.1 Data

To examine the presence of time variation in credit uncertainty, we begin by looking at the dynamics of credit aggregates. Figure 1 plots the quarterly growth rate of credit available to the private nonfinancial sector in the United States from 1979Q1 to 2019Q4; this includes nonfinancial corporations (both private-owned and public-owned), households, and nonprofit institutions serving households as defined in the System of National Accounts 2008. In terms of financial instruments, credit covers loans and debt securities.[5] We start our analysis with this broad measure of credit and use this definition when discussing the results in Sections 2, 3, and 4. Section S4

5. For more details, see credit data on the BIS (Bank for International Settlements) statistics repository. The series source code is Q: U.S.:P:A:M:USD:A.

of the online supplement carries out robustness checks accounting for different measures of credit aggregates and a longer sample. It that the estimates characterizing credit uncertainty is a robust empirical feature across these different measures and sample choices.

1.2 Empirical Model

The time variation in credit uncertainty is isolated by considering a stochastic volatility (SV) model for the data y_t, allowing for correlation between shocks to the level and the log-volatility of credit growth rate (leverage effect). The observation equation is

$$y_t = \phi_y y_{t-1} + \exp(h_t/2)\,\varepsilon_t, \quad \varepsilon_t \sim N(0,1), \quad t = 1, 2, \cdots, T. \tag{1}$$

The log-volatility h_t follows:

$$h_{t+1} = \mu_h + \phi_h(h_t - \mu_h) + \tau\eta_{t+1}, \quad \eta_{t+1} \sim N(0,1), \quad t \geq 1, \tag{2}$$

$$\begin{pmatrix} \epsilon_t \\ \eta_{t+1} \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}\right). \tag{3}$$

The parameters μ_h and τ control the level of mean volatility and the extent of SV in y_t, respectively. The process $\{y_t\}$ is hit by both ϵ_t and η_t; the innovation ϵ_t to the observation y_t changes the level of y_t; the innovation η_t to the volatility affects the standard deviation of y_t. When estimating the model, the data are demeaned. Therefore, uncertainty shocks in this environment are captured by η_t (the stochastic disturbance to the latent volatility process).

The parameter ρ controls the strength of the dependence between ϵ_t and η_{t+1}—the size of the "leverage effect" of the level shock ϵ_t on the volatility shock η_{t+1}. The standard SV model corresponds to $\rho = 0$. In the estimation exercise, we allow for potential correlation between ϵ_t and η_{t+1} and estimate the SV model with and without the leverage effect.

1.3 Bayesian Estimation

Overview. This section discusses the Bayesian estimation of the SV model in Section 1.2. For Bayesian inference, the joint posterior distribution of θ and $h_{1:T}$ is

$$p(\theta, h_{1:T}|y_{1:T}) = \frac{p(h_{1:T}, y_{1:T}|\theta)p(\theta)}{p(y_{1:T})}, \tag{4}$$

$$p(h_{1:T}, y_{1:T}|\theta) = p(h_1|\theta)p(y_1|h_1,\theta)p(h_2|h_1,y_1,\theta)p(y_2|h_2,y_1,\theta)\times$$

$$\prod_{t=3}^{T} p(h_t|h_{t-1}, y_{t-1}, y_{t-2}, \theta)p(y_t|h_t, y_{t-1}, \theta), \quad (5)$$

$p(\theta)$ is the prior density for θ, and

$$p(y_{1:T}|\theta) = \int p(y_{1:T}|h_{1:T}, \theta)p(h_{1:T}|\theta)dh_{1:T} \quad (6)$$

is the likelihood. The details about each density in equation (5) are given in Section S1.3 of the online supplement. The vector of model parameters for the univariate SV model with leverage is $\{\mu_h, \phi_y, \phi_h, \tau, \rho\}$.

Choice of priors. We follow Kim, Shephard, and Chib (1998) and choose the prior for the persistence parameters ϕ_y and ϕ_h as $(\phi + 1)/2 \sim \text{Beta}(a_0 = 20, b_0 = 1.5)$, that is,

$$p(\phi) = \frac{1}{2B(a_0, b_0)}\left(\frac{1+\phi}{2}\right)^{a_0-1}\left(\frac{1-\phi}{2}\right)^{b_0-1}. \quad (7)$$

The prior for τ is the half-Cauchy distribution, that is, $p(\tau) = (2I(\tau > 0))/(\pi(1+\tau^2))$, the prior for $p(\mu_h) \propto 1$, and the prior for $p(\rho) \sim U(-1, 1)$. These prior densities cover most possible values in practice and are noninformative and independent.

Estimating likelihood. The likelihood function for the univariate SV model in equation (6) is computationally intractable because it is a high-dimensional integral over the latent states $h_{1:T}$. The correlated pseudo-marginal Metropolis–Hastings (PMMH) algorthim of Deligiannidis, Doucet, and Pitt (2018) is used to estimate the SV model. It is run for 15,000 iterations, with the initial 5,000 iterations discarded as burn-in. We use an adaptive random walk proposal -- $q\left(\theta^{'}|\theta\right)$ and, following Garthwaite, Fan, and Sisson (2016), adaptively tune the scaling factor of the covariance matrix. This enables us to prespecify the overall acceptance probability before running the correlated PMMH method. In the examples, the overall acceptance probability is set at 25% and the number of particles to 100. Section S1 of the online supplement discusses the estimation method for the SV model.

2. EMPIRICAL RESULTS

Table 1 reports the posterior mean estimates of the SV parameters for the standard SV model without leverage. Table S1 of Section S3 of the online supplement reports the results for the SV model with leverage. In the estimation, we find weak evidence supporting the significance of the parameter ρ describing the leverage effect; for the remaining parameters, the estimates are comparable between the two specifications. For calibrating the theoretical model in Section 3, therefore, we use estimated

TABLE 1

Posterior Mean Estimates (with 95% Credible Intervals in Brackets) of the Standard SV Model without Leverage

Parameters	Priors	SV
μ_h	Flat prior: $p(\mu_h) \propto 1$	-10.27 $(-10.84, -9.7)$
ϕ_h	Beta(20,1.5)	0.867 (0.62,0.97)
ϕ_y	Beta(20,1.5)	0.836 (0.73,0.93)
τ	HC(0,1)	0.359 (0.15,0.72)

parameters presented in Table 1. The table shows that for the standard SV model: (i) the average volatility of an innovation to credit growth rate, μ_h, is large, revealing a large degree of volatility in credit growth rate and the posterior of μ_h is tightly concentrated; (ii) there is substantial evidence for SV in the data for credit growth rate (a large τ); (iii) the shocks to the level and log-volatility of total credit are quite persistent (large ϕ_y and ϕ_h).[6]

We now try to understand how well our estimated measure of latent volatility captures uncertainty in financial markets. Figure 2 plots the posterior mean estimate of the latent volatility and the one-step-ahead financial uncertainty index (LMN) constructed in Ludvigson, Ma, and Ng (2021). The LMN financial uncertainty index is constructed using information from 148 monthly financial indicators. We use a quarterly average of the LMN index and standardize the measures to make them comparable. Figure 2 shows that our estimated measure of uncertainty described by the latent volatility underlying the credit growth rate does a reasonable job of tracking aggregate uncertainty in financial markets compared to the more informative LMN index. Similar to the LMN index, our measure records elevated levels during downturns in the business cycle. Unlike the LMN index, which spiked during financial crisis episodes in 1987 and 1989, our measure of latent volatility remained elevated throughout the mid-1980s. This period also coincides with banking deregulation.

Figure 2 suggests that using credit aggregates in constructing measures of uncertainty helps describe aggregate uncertainty in financial markets. However, isolating the sources of variation in estimated aggregate uncertainty, that is, whether the disturbances originate due to changes in supply or demand, is challenging. We next examine the potential relation between the estimated shocks to the latent volatility underlying credit growth rate, the tightening of credit supply conditions and conditions affecting credit demand.

6. In addition to considering the broad measure of credit growth rate, we also consider additional measures that examine the growth rate in credit extended specifically to households and firms. The findings are consistent with Table 1. Detailed results are in Table S2 in Section S4 of the online supplement. When we examine the real effects of shocks to credit uncertainty in Section 4, we augment the observation equation with variables that are most useful for predicting credit growth rate to remove any additional predictability. The estimates are comparable across both specifications.

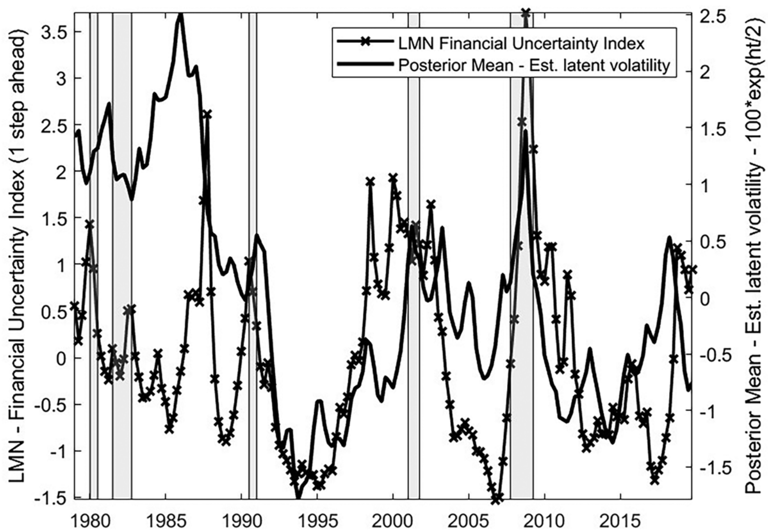

Fig 2. Time-Varying Volatility in Credit Growth Rate.

NOTES: The standardized posterior mean estimate of the latent volatility underlying the credit growth rate, calculated as $100\exp(h_t/2)$, (solid black line), and the standardized measure of the one-step-ahead LMN financial uncertainty index (crossed black line) . Shaded gray areas show NBER recessions.

To carry out this exercise, we use the Senior Loan Officer Opinion Survey (SLOOS) on Bank Lending Practice,[7] which gives the net percentage of respondents reporting *Tightening Standards* for C&I Loans (Commercial and Industrial) for large and medium-sized entities and the net percentage of respondents reporting *Stronger Demand* for C&I Loans for large and medium-sized entities. The former captures the supply side, and the latter describes the demand side. The SLOOS is available for different measures of credit activity; however, the time series for C&I loans is the longest.[8]

Figure 3 plots the net percentage of domestic respondents reporting tightening standards for C&I loans (crossed black line) and the net percentage of domestic respondents reporting stronger demand for C&I loans (dashed black line) along with the extracted shocks to latent volatility (solid black line).[9]

7. The data from the Senior Loan Officer Opinion Survey are obtained from information published on the website of the Board of Governors of the Federal Reserve System. For more details, see https://www.federalreserve.gov/data/sloos/sloos-202004-chart-data.htm.

8. The figure and analysis can also be replicated using SLOOS responses for C&I loans to small entities. The correlation between the SLOOS responses for large and medium-sized entities and SLOOS survey responses for small entities is very high—0.96 for survey responses corresponding to questions relating to credit supply and 0.92 for survey responses corresponding to questions relating to credit demand.

9. The SLOOS data for credit supply conditions are available since 1990Q2 and for credit demand conditions since 1991Q4.

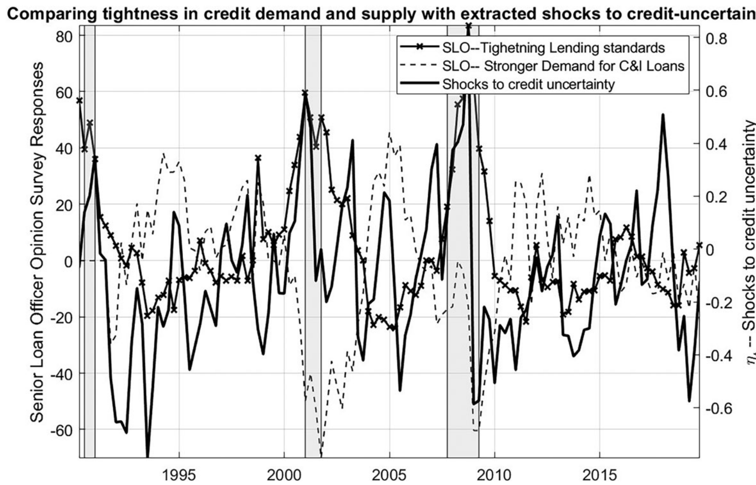

Fig 3. Estimated Shocks and Changes in Conditions Underlying Credit Supply and Demand.

NOTES: Credit Supply: Senior Loan Officer Opinion Survey: Net Percentage of Domestic Respondents Reporting Tightening Standards for C&I Loans (crossed black line); Credit Demand: Senior Loan Officer Opinion Survey: Net Percentage of Domestic Respondents Reporting Stronger Demand for C&I Loans (dashed black line); estimated shocks to credit uncertainty (solid black line). Shaded gray areas show NBER recessions.

TABLE 2

REPORTING CORRELATION COEFFICIENTS, F -STATISTIC, AND p -VALUES OBTAINED BY REGRESSING THE ESTIMATED SHOCKS TO CREDIT UNCERTAINTY ON SURVEY RESPONSES FROM THE SLOOS

Variable	Raw correlation	R^2	F-statistic	p-Value
Net Percentage of Domestic Respondents Reporting Tightening Standards for C&I Loans	0.30	0.0901	11.1870	0.0011
Net Percentage of Domestic Respondents Reporting Stronger Demand for C&I Loans	−0.0452	0.0020	0.2186	0.6411

This suggests that our measure is possibly more related to changes in credit supply. Table 2 reports the raw correlation coefficient between the proxies capturing tightening (strengthening) in credit supply (demand) and estimated shocks to credit uncertainty along with the F-statistic for the R^2 measure.

The Senior Loan Officer survey explicitly asks respondents if changes in lending standards are affected by the less favorable or more uncertain economic outlook. One interesting pattern in the survey is that uncertainty in economic outlook becomes very

TABLE 3

LEAD/LAG CORRELATION: CORR $(\sigma_{h,t}, \text{VAR}_{x,t+k})$, WHERE $\sigma_{h,t} = 100\exp(h_t/2)$ IS OUR MEASURE, $\text{VAR}_{x,t+k}$ ARE THE ALTERNATIVE MEASURES OF UNCERTAINTY. k IS THE NUMBER OF QUARTERS

Alternative measures of uncertainty	k	−3	−2	−1	0	1	2	3	4
VIX		0.19	0.22	0.24	0.31	0.35	0.37	0.42	0.44
BBD – Economic Policy Uncertainty Index		0.21	0.23	0.24	0.29	0.27	0.25	0.27	0.31
JLN – Macro Uncertainty Index/$h = 1$		0.37	0.40	0.43	0.48	0.52	0.53	0.54	0.53
JLN – Real Uncertainty Index/$h = 1$		0.29	0.31	0.34	0.38	0.40	0.42	0.42	0.42

important in tightening lending standards at the onset of recessions. In contrast, the opposite is true during booms.

The estimated shocks to credit uncertainty reflect this—the upward spike in credit uncertainty in recessions is likely a consequence of a tightening credit supply. The evidence, therefore, suggests a link between unforeseen changes in credit uncertainty and tighter credit supply conditions.

Finally, to understand how our measure compares with existing measures of uncertainty, Table 3 reports the correlation at various leads and lags for our measure of credit uncertainty and commonly used measures of uncertainty (VIX, JLN Index: Jurado, Ludvigson, and Ng (2015), BBD Index: Baker, Bloom, and Davis (2016)). While a positive correlation is observed, it is important to note that the estimate does not move one-for-one with these measures.

The following section develops a theoretical model explaining how changes in credit uncertainty can impact real activity, highlighting the underlying transmission mechanism.

THEORETICAL MODEL

We use a model of collateral constraints in the spirit of Kiyotaki and Moore (1997) as a theoretical framework for interpreting the empirical results. In a standard RBC model, a role for credit uncertainty is introduced by incorporating a working capital requirement on the firms' side. Credit uncertainty is modeled as the time variation in the volatility of credit available for financing working capital requirements. We assume that firms must pay labor at the beginning of each period before producing output. Firms finance this labor payment with an intratemporal loan (similar to Carlstrom and Fuerst 1997). There is no interest rate on financing the working capital because the firms pay off the loans within the period. However, the amount a firm can borrow is limited to a fraction of the value of firm capital. The specification of the model environment follows Sims (2017). We abstract from intertemporal debt in the model as the intuition governing the impact of an uncertainty shock remains unchanged compared to the scenario with intertemporal debt. Having a simpler specification also allows us to decompose the direct effects of uncertainty from the indirect effects that come from the higher-order terms in Section 3.4.[10]

2.1 Model Description

This is a model in discrete time where agents live infinitely. Households consume (C_t), supply labor (N_t), and save (B_t); R_t is the gross rate of return on savings. Household utility is of CRRA type, additively separable in labor and consumption; γ_c is the inverse of the intertemporal elasticity of substitution, χ is the inverse of the Frisch elasticity, β is the discount factor, and θ is set such that in steady state, the hours supplied is one-third. Households solve

$$\max_{\{C_t,N_t,B_{t+1}\}} E_0 \sum_{t=0}^{\infty} \beta^t \left(\frac{C_t^{1-\gamma_c}}{1-\gamma_c} - \theta \frac{N_t^{1+\chi}}{1+\chi} \right)$$

subject to

$$C_t + B_{t+1} = W_t N_t + R_{t-1} B_t + \Pi_t,$$

where W_t is the real wage and Π_t denotes firms' residual profits. The first-order conditions of the household are

$$C_t^{-\gamma_c} = \lambda_t, \tag{8}$$

$$\theta N_t^{\chi} = \lambda_t W_t, \tag{9}$$

$$1 = \beta E_t \left[\left(\frac{C_{t+1}}{C_t} \right)^{-\gamma_c} R_t \right]. \tag{10}$$

The firms own capital stock (K_t) and hire labor input (N_t) from households to produce output Y_t using the Cobb–Douglas production technology

$$Y_t = A_t K_t^{\alpha} N_t^{1-\alpha}, \tag{11}$$

where A_t describes the state of aggregate technology. Capital accumulation is subject to adjustment costs and evolves as

$$K_{t+1} = (1-\delta)K_t + I_t - \frac{\phi}{2} \left(\frac{I_t}{K_t} - \delta \right)^2 K_t, \tag{12}$$

10. Section S5 of the online supplement considers a richer environment, including intertemporal debt and debt financing both the wage bill and investment. The intuition underlying the transmission mechanism is unchanged, and the results are qualitatively comparable.

where I_t is investment, δ is the rate at which capital depreciates, and ϕ is the cost of adjusting capital. In this framework, investment is financed using residual profits (dividends). The dividend payout is

$$\Pi_t = Y_t - W_t N_t - I_t.$$

We introduce a working capital constraint in this simple RBC setup and assume that firms must pay labor prior to producing output, and the working capital is financed through an intratemporal loan.[11] The constraint restricting working capital financing is

$$W_t N_t \leq \zeta_t q_t K_t. \tag{13}$$

The amount the firm can borrow, as seen from equation (13), is limited to a fraction ζ_t of the total value of capital, where q_t is the shadow price of capital and is defined using the Lagrange multiplier on the law of motion of capital in the optimization problem of the firm that is discussed below. The firm chooses investment (I_t), capital (K_{t+1}), and labor (N_t) by maximizing the discounted sum of dividends (Π_t)

$$\max_{\{I_t, N_t, K_{t+1}\}} E_0 \sum_{t=0}^{\infty} \beta^t \frac{\lambda_t}{\lambda_0} \left[A_t K_t^{\alpha} N_t^{1-\alpha} - W_t N_t - I_t \right]$$

subject to the law of motion for capital—equation (12) and the constraint restricting working capital financing—equation (13), respectively. Firm dividends at time $t + j$ are discounted using the stochastic discount factor from the optimization problem of the households—$\beta^j \lambda_{t+j}/\lambda_t$.

Let μ_t be the multiplier on the working capital constraint; $\mu_t > 0$ implies that the constraint binds and changes in ζ_t impact the equilibrium conditions of the model. The first-order conditions corresponding to the optimal choices of labor (N_t), investment (I_t), and capital (K_{t+1}) are

$$W_t(1 + \mu_t) = (1 - \alpha) A_t K_t^{\alpha} N_t^{-\alpha}, \tag{14}$$

$$1 = q_t \left[1 - \phi \left(\frac{I_t}{K_t} - \delta \right) \right], \tag{15}$$

$$q_t = \beta E_t \frac{\lambda_{t+1}}{\lambda_t} \left[\alpha A_{t+1} K_{t+1}^{\alpha-1} N_{t+1}^{1-\alpha} + q_{t+1} \left((1 - \delta) + \mu_{t+1} \zeta_{t+1} - \frac{\phi}{2} \left(\frac{I_{t+1}}{K_{t+1}} - \delta \right)^2 \right. \right.$$

11. While we work with a reduced-form specification in equation (13), in Section S6 of the online supplement, we provide microfoundations to equation (13) that incorporate a "skin in the game" constraint in the optimal contracting problem.

$$+ \ \phi\left(\frac{I_{t+1}}{K_{t+1}} - \delta\right)\frac{I_{t+1}}{K_{t+1}}\Bigg)\Bigg], \tag{16}$$

and

$$\mu_t(W_tN_t - \zeta_t q_t K_t) = 0, \ \text{with } \mu_t \geq 0. \tag{17}$$

When the credit constraint binds, $\mu_t > 0$ and $(W_tN_t - \zeta_t q_t K_t) = 0$; when the constraint is slack, $\mu_t = 0$ and $(W_tN_t - \zeta_t q_t K_t) < 0$. The market clearing condition is

$$Y_t = C_t + I_t. \tag{18}$$

Credit uncertainty: Credit uncertainty in the model is specified as follows. We assume that ζ_t—the fraction of the value of capital against which the firm that can borrow (equivalently, the loan-to-value ratio), follows an AR(1) process with a time-varying second moment,

$$\begin{aligned} \zeta_t - \zeta_{ss} &= \rho(\zeta_{t-1} - \zeta_{ss}) + \exp(h_t/2)\epsilon_t, \\ h_t &= (1 - \rho_h)\overline{h} + \rho_h h_{t-1} + \tau\eta_t, \end{aligned} \tag{19}$$

where ζ_{ss} is the fraction the firm can borrow against the value of capital in steady state; $\overline{h}$ is the average uncertainty in credit availability, and τ is the extent of stochastic volatility (SV) in credit availability. In DSGE models, the loan-to-value ratio is often used as a proxy to capture the average stance of credit availability in the economy, that is, a supply-side measure (see, for instance, Jensen, Petrella, Ravn, and Santoro 2020). Our interpretation is similar and consistent with the results from Section 2 where we show that the estimated shocks comove more strongly with indicators of credit supply than credit demand. *Ceteris paribus*, a shock to the first moment of ζ_t (ϵ_t) in the model relaxes the borrowing constraint, and firms have access to more credit. Likewise, a shock to the second moment of ζ_t (η_t) generates uncertainty about credit available in the economy.

The evolution of SV in the model is identical to that estimated in Section 2 and is similar to that presented in Fernández-Villaverde et al. (2015) (, and Fernández-Villaverde and Guerrón-Quintana (2020).[12] Equations (8)–(19) summarize the equilibrium conditions in the model.

2.2 *Model Solution*

This paper aims to explore the effects of a change in the second moment that captures shocks to credit uncertainty. A first-order approximation shuts down the effects of changes in higher-order moments by construction. A second-order-accurate so-

12. Our results are robust to the alternative specification in Basu and Bundick (2017).

lution impacts expected values but does not influence the dynamics as higher-order terms do not independently enter the solution (Schmitt-Grohé and Uribe 2004). It is, therefore, necessary to consider a third-order approximation for uncertainty to have dynamic effects. We do so by using the perturbation methods combined with pruning (to prevent explosive solutions) in Andreasen, Fernández-Villaverde, and Rubio-Ramírez (2018). In solving the model, we also need to consider the occasionally binding working capital constraint. Existing methods of solving models with occasionally binding constraints, such as Guerrieri and Iacoviello (2015), apply it in the context of models solved using a first-order approximation. Our model includes uncertainty shocks, and to simplify the solution, we assume that the credit constraint always binds.[13]

2.3 Model Calibration

The model is calibrated quarterly, and the behavioral parameters of the model are standard. The share of capital in the Cobb–Douglas production function is set at $\alpha = 0.33$, the discount factor $\beta = 0.99$, the depreciation rate of capital $\delta = 0.02$, and investment adjustment costs $\phi = 4$. The parameter θ scaling the disutility of labor supply is set such that that labor hours in steady state are $\frac{1}{3}$. The inverse intertemporal elasticity of substitution is fixed at $\gamma_c = 1$ and the inverse of the Frisch elasticity of labor supply at $\chi = 1$. We set the state of aggregate technology A_t to 1.

We next calibrate the steady-state value of ζ_t and the parameters governing the evolution of ζ_t. Note that when the collateral constraint $\mu > 0$ binds in steady state, it imposes certain restrictions on the steady-state value of ζ_t given by ζ_{ss}. For $\mu > 0$, in steady state, it can be shown that $\zeta_{ss} < \frac{1-\alpha}{\alpha}\left(\frac{1}{\beta} - (1-\delta)\right)$. Given the values of α, β, δ, this implies that $\zeta_{ss} < 0.0602$ (see Section S8 of the online supplement for the derivation). The assumption of a binding constraint thus imposes an upper limit on the steady-state value of ζ_t. For the first part of the analysis that examines the transmission mechanism, ζ_{ss} is set to 3.5%. Later on, in understanding the differential effects of changes in credit uncertainty (Section 3.2) in recessions and expansions, we relax this assumption and solve the fully nonlinear model using global methods, allowing occasionally binding constraints.

Finally, we consider calibrating the remaining parameters characterizing the persistence and standard deviation of shocks to the first and second moment of ζ_t. In the model, ζ_t is interpreted to capture the average stance of credit available in the

13. To check the accuracy of this simplified approach, we solve the fully nonlinear model, allowing for occasionally binding constraints using global methods. The results are robust to the simplifying assumption that the constraint always binds. The solution method involving a third-order approximation around a constrained binding steady state is significantly faster than solving the global model. This also allows us to decompose the response on impact into the direct effect that operates via the precautionary response and the indirect effect that operates via the restrictions imposed by the equilibrium conditions of the model. However, when examining the state-dependent effects of shocks to credit uncertainty in Section 4.2, we solve the fully nonlinear model, allowing for occasionally binding constraints. Section S7 of the online supplement presents the details of the solution technique and the results from solving the fully nonlinear model, allowing the credit constraint to bind occasionally.

economy. Therefore, we use the posterior mean estimates from Table 1 to calibrate these parameters.

2.4 Transmission of Shock to Credit Uncertainty

Before describing the mechanism guiding the transmission of shocks to credit uncertainty in the model, we discuss equilibrium conditions that play an important role in generating business cycle comovement in macro variables.

The behavior of the firm in this environment is subject to collateral constraints on borrowing. Since the constraint limits the quantity a firm may borrow, it has implications for how much labor the firm can hire. The collateral constraint on working capital thus introduces a wedge between the frictionless competitive equilibrium and the realized outcome in the constrained economy. This simple assumption generates important implications by introducing a distortion between labor demanded by firms and labor supplied by households.

Why is this feature important? Basu and Bundick (2017) show that in a flexible-price model, an uncertainty shock, introduced through time variation in the volatility of household preferences (and aggregate technology), cannot generate a simultaneous decline in consumption, investment, and output. The feature of simultaneous decline is regarded as a stylized fact that characterizes the effect of an uncertainty shock. Basu and Bundick (2017) subsequently introduce nominal rigidities and highlight the role of the f markup (distorting the flexible price equilibrium) in determining labor demand as the key feature that delivers this simultaneous decline. In our environment, however, uncertainty is introduced through a credit channel. The collateral constraint in our model, operating through the intratemporal working capital channel, distorts the flexible-price equilibrium. The labor demand side of the model now reflects the effect of the collateral constraint, with

$$W_t(1+\mu_t) = (1-\alpha)A_t K_t^{\alpha} N_t^{-\alpha},$$

where, the binding constraint on working capital.is given by

$$W_t N_t = \zeta_t q_t K_t.$$

The collateral constraint thus introduces this labor wedge – $(1+\mu_t)$ that distorts equilibrium conditions describing labor demand while the labor supply side remains unaffected by this distortion;

$$W_t = \frac{\theta N_t^{\chi}}{C_t^{-\gamma_c}}.$$

The labor wedge, along with the binding collateral constraint, thus provides the additional margin that can dampen labor demand in response to an increase in credit uncertainty, which is a key element in generating business cycle comovement.

Figure 4 presents the impulse responses of variables to a one-standard-deviation increase in credit uncertainty. An uncertainty shock in the model triggers a

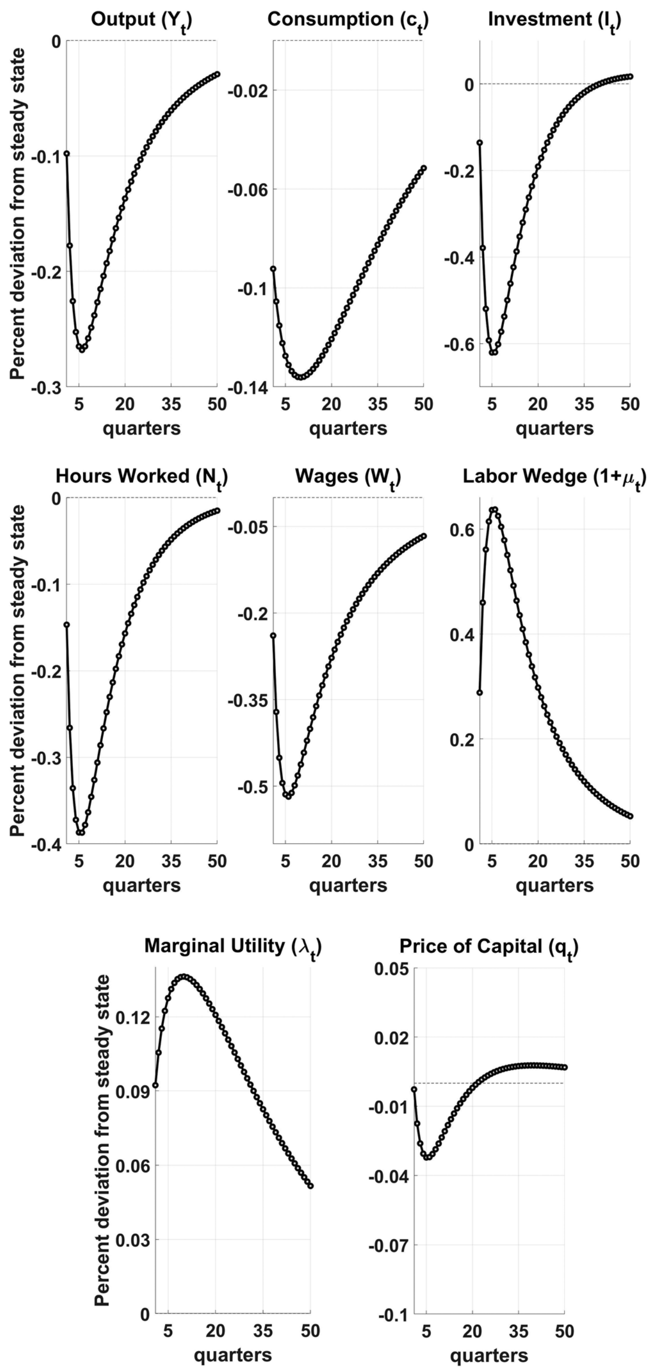

Fig 4. Impulse Responses to a One-Standard-Deviation Increase in Credit Uncertainty in the Theoretical Model.

NOTES: A shock to credit uncertainty triggers a simultaneous decline in output, consumption, investment, hours, and real wages.

precautionary savings response in households, leading to a decrease in consumption and an increase in the marginal utility and labor supply. This household response is expansionary for hours and investment while contractionary for consumption. However, with collateral constraints, an increase in credit uncertainty triggers an increase in the labor wedge $(1 + \mu_t)$, which generates a contraction in labor demand.

Why does the labor wedge, $(1 + \mu_t)$, increase in response to a shock to credit uncertainty? When the multiplier on the borrowing constraint is relaxed (via an increase in μ_t), it quantifies an increase in the wage bill. In response to an increase in uncertainty about credit availability—firms are uncertain about how much credit they can get to finance working capital, that is, labor payments. This increases perceived labor costs and is reflected via increases in the labor wedge, causing a decline in labor demand.

If the decline in labor demand exceeds the precautionary-induced expansion in labor supply, then, in equilibrium, a shock to credit uncertainty will generate a fall in hours along with a decline in real wages. An uncertainty shock also reduces the price of capital, which drives the fall in investment. Given this is a closed economy model, when theese effects are combined, a shock to credit uncertainty in a flexible-price model with collateral constraints can generate a sizable and simultaneous decline in hours, real wages, output, consumption, and investment.

To better understand how the precautionary motives interact with the endogenous variables in generating business cycle comovement, we use the pruned third-order solution and decompose the effect on impact into two channels—one that captures the *direct effect* and the other capturing the *indirect effect.*

The direct effect can be interpreted to capture the adjustment to uncertainty, which is linear in the states (as shown in Andreasen 2012) and quantifies the precautionary component of the response. The indirect effect can be interpreted to capture the endogenous transmission occurring through the restrictions imposed by the equilibrium conditions of the model. This decomposition allows us to disentangle the relative strengths of the expansionary effects vis-à-vis the contractionary effects. The response on impact is a sum of these two effects and corresponds to the response on impact in Figure 4, discussed above. Table 4 summarizes the results of this decomposition. Details of the derivation can be found in Section S9 of the online supplement.

Table 4 shows that the direct effect of an uncertainty shock triggers an increase in the marginal utility and the price of capital, along with a decline in the rate of return on capital and consumption. It is expansionary for hours and investment while contractionary for consumption. The direct effect on impact also increases the labor wedge, distorting labor demand.

In addition to the direct effect, there are indirect effects operating through the higher-order interaction between the endogenous variables and the uncertainty shock in the model. Table 4 shows that the indirect effects arising from the interaction between the shock and the higher-order terms dampen total hours supplied, price of capital, GDP, and investment but amplify the initial decline in consumption from the precautionary response. The indirect effect also reinforces the increase in labor wedge. As seen from Table 4, the increase in labor wedge via the direct effect is

TABLE 4

Decomposing the Effects of a One-Standard-Deviation Increase in Credit Uncertainty

Variables	Direct effect (% change) (1)	Indirect effect (% change) (2)	Total effect on impact (% change) (3)
Consumption	−0.065	−0.028	−0.092
Real wages	−0.023	−0.216	−0.239
Rate of return on capital	−0.029	0.003	−0.026
GDP	0.028	−0.126	−0.098
Hours	0.042	−0.189	−0.147
Marginal utility	0.065	0.028	0.092
Labor wedge	0.009	0.279	0.288
Price of capital	0.019	−0.022	−0.003
Investment	0.236	−0.372	−0.136

Note: The direct effects show that an increase in credit uncertainty is contractionary for consumption but expansionary for hours and investment. Direct effects also increase the labor wedge and the price of capital. The indirect effects reinforce the direct effects for consumption and labor wedge and counter the direct effect for investment, hours, real wages, and the price of capital. The total effect on impact is a sum of the direct and indirect effects.

insufficient to deliver a contraction in hours. However, once combined with the indirect effect, the decline in labor demand due to the increase in labor wedge is strong enough to offset the increase in labor supply.

Along with the response of the labor wedge, the response of the price of capital is also important toward generating business cycle comovement. Similarly to the response of the labor wedge, the interaction effect generates a contraction in the price of capital that more than offsets the increase due to the direct effect. This decline in the price of capital generates a fall in investment. Our mechanism operating through the labor wedge, which depends on the Lagrange multiplier on the collateral constraint and the price of capital, is similar to the intuition described in Liu, Wang, and Zha (2013), who examine the implication of a shock to the first moment of the time-varying collateral constraint. However, instead of transmitting through first-moment changes in the marginal financing condition, our results propagate via a third-order precautionary response accompanied by a nonlinear interaction between the shock and the endogenous variables.

These results operating through the firm's binding borrowing constraint are insightful. An uncertainty shock of this form generates a simultaneous decline in hours, real wages, consumption, investment, and output in a parsimoniously specified environment. One of the challenges in the theoretical literature examining the effects of uncertainty has been to generate a sizable and simultaneous impact for endogenous variables in response to changes in the second moment. Our proposed shock and model with a collateral constraint can significantly impact macro-economic variables in response to changes in credit uncertainty.

The next section tests if the theoretical predictions of the model can be recovered in the data. Thus, we empirically test if shocks to credit uncertainty generate patterns suggested by the model.

3. REAL EFFECTS OF CHANGES IN CREDIT UNCERTAINTY IN THE DATA

Equation (1) assumes an AR(1) structure underlying the evolution of credit growth rate. This specification aligns with how we model credit uncertainty in the DSGE model. When empirically examining the real effects of unforeseen changes in credit uncertainty, we consider a richer structure for the observation equation with

$$y_t = \sum_{j=1}^{4} \phi_{y,j} y_{t-j} + \phi_{\Delta} \Delta GDP_t + \phi_c c_{t-1} + \exp(h_t/2)\epsilon_t, \quad \epsilon_t \sim N(0,1), \quad t = 1, 2, ., T, \tag{20}$$

and

$$h_{t+1} = \mu_h + \phi_h (h_t - \mu_h) + \tau \eta_{t+1}, \quad \eta_{t+1} \sim N(0,1), \quad t \geq 1. \tag{21}$$

The evolution of the log volatility underlying credit growth rate (h_t) in equation (21) is unchanged; however, the observation equation is modified to include four lags of credit growth rate, contemporaneous GDP growth rate (ΔGDP_t) along with a lagged value of the responses from the SLOOS capturing expectations about credit conditions—Net Percentage of Domestic Banks Reporting Increased Willingness to Make Consumer Installment Loans (c_{t-1}).[14] These controls eliminate the possibility of additional predictable variation in credit growth rate, and help identify true surprises to credit uncertainty in the data. Table 5 presents the estimates for the updated and baseline specifications in Section 1.2 and shows that the estimates describing the evolution of log volatility remain comparable after controlling for the possibility of additional predictable variation in credit growth rate. If anything, the estimate underlying the extent of SV -- τ is larger.

Using estimated values of η_t from the system in equations (20) and (21), we estimate the real effects of changes in credit uncertainty with local projections following Jordà, Schularick, and Taylor (2013). Within the framework of a DSGE model, it is straightforward to isolate the effects of uncertainty shocks. However, isolating the change in uncertainty exogenous to macro-economic conditions in an empirical environment is more challenging. To circumvent this challenge, a rich set of controls is included in constructing local projections that allow us to extract responses in macro variables due to changes in credit uncertainty. The description of controls follows later. Equation (22) describes our specification for constructing the regime-independent responses of macro-economic variables at horizon h for changes in credit uncertainty in period t,

$$x_{t+h} = \alpha_h + \psi_h(L) z_t + \beta_h \eta_t + \epsilon_{t+h}, \tag{22}$$

14. The data can be accessed at https://fred.stlouisfed.org. Variable id: DRIWCIL. This variable proxies the supply side and is an important predictor of the credit growth rate in the economy.

TABLE 5

SUMMARIZING ESTIMATES ACROSS DIFFERENT MODEL SPECIFICATIONS

	Model 1 Q1-78/Q4-19	Model 2 Q1-78/Q4-19
μ_h	−10.27 (−10.84, −9.7)	−10.92 (−11.8, −10.20)
ϕ_h	0.867 (0.62,0.97)	0.84 (0.63,0.97)
$\phi_{y,1}$	0.836 (0.73,0.93)	0.186 (0.00,0.36)
$\phi_{y,2}$	NA	0.358 (0.20,0.52)
$\phi_{y,3}$	NA	0.162 (−0.001,0.32)
$\phi_{y,4}$	NA	0.253 (0.094,0.42)
$\phi_{\Delta_{GDP}}$	NA	0.062 (−0.04,0.16)
ϕ_c	NA	0.008 (0.001,0.02)
τ	0.359 (0.15,0.72)	0.473 (0.23,0.80)

NOTE: Model 1 is the baseline, including only one lag of credit growth rate in the measurement equation, and Model 2 includes contemporaneous change in GDP, additional lags of credit growth rate, and lagged responses from the Senior Loan Officer Opinion Survey (SLOOS).

where x is the macro-economic variable of interest, z is a vector of control variables (includes contemporaneous and lagged variables), $\psi_h(L)$ is a second-order polynomial in the lag operator, and η_t is the unforeseen change in estimated latent volatility from the model presented in Table 5. In the linear model, the coefficient of interest at each horizon is β_h.

We also consider state dependence in computing local projections to distinguish between the effects of an increase in credit uncertainty in recessions and expansions. To introduce state dependence in quantifying the real effects of changes in credit uncertainty, we follow Ramey and Zubairy (2018) and extend the linear model in equation (22) to (23) that allows for state dependence in calculating impulse responses

$$x_{t+h} = I_{t-1}[\alpha_{R,h} + \psi_{R,h}(L)z_t + \beta_{R,h}\eta_t + \epsilon_{t+h}] + (1 - I_{t-1})[\alpha_{NR,h} + \psi_{NR,h}(L)z_t + \beta_{NR,h}\eta_t + \epsilon_{t+h}], \quad (23)$$

where I_{t-1} is a dummy variable indicating whether the economy is in a recession when the shock hits.[15] All the model coefficients are allowed to vary according to the state of the economy. To account for the serial correlation in the error terms, we use the Newey–West correction for calculating standard errors (Newey and West 1987). In the state-dependent extension, the coefficients of interest at each horizon are $\beta_{R,h}$ (recessions) and $\beta_{NR,h}$ (expansions), respectively.

15. Recessions are defined using the NBER business cycle chronology.

Control variables: As highlighted earlier, it is important to include a sufficient set of controls to disentangle the effects of credit uncertainty on the path of real activity. The set of controls used in constructing local projections comprises indicators describing the state of financial markets outside of our measure, the stance of monetary policy, an economic uncertainty index, and lags of relevant macro-economic variables.

To control for the state of financial conditions, we use lags of a financial stress index (the Chicago Fed's National Financial Conditions Index (NFCI)). The NFCI published by the Chicago FED and developed in Brave and Butters (2012), is an index constructed using over 100 variables describing conditions related to risk, credit, and leverage in financial markets. The inclusion of the NFCI controls for conditions that affect financial markets but are not captured by the dynamics of credit aggregates alone. Treasury yields on the 2-year constant maturity Treasury Security is used to control the stance of monetary policy.[16] We control for changes in aggregate macro-economic uncertainty by including the one-step-ahead real economic uncertainty index constructed in Jurado, Ludvigson, and Ng (2015). Finally, we include lags of GDP growth rate and lags of the dependent variable as well.

Measures of economic activity: In the theoretical model, we examine the impulse responses of macro-economic aggregates. We expand this set to include more detailed measures in constructing local projections. We thus construct local projections for a wide range of macro-economic activity, including credit-dependent measures of real activity (total vehicle sales and durable consumption) and broader measures of macro-economic activity (aggregate consumption, nondurable consumption, investment, GDP, unemployment rate, and total employed). Local projections are computed using growth rates (log first differences) of total vehicle sales, aggregate consumption, durable consumption, nondurable consumption, investment, GDP, and nonfarm employment. The shocks are standardized such that the coefficients in equations (22) and (23) can be interpreted as the elasticity of the relevant variable with respect to a one-standard-deviation increase in uncertainty.

3.1 Impact of Unforeseen Increase in Credit Uncertainty

We begin by highlighting the importance of state dependence in quantifying the effects of an unforeseen increase in credit uncertainty by first comparing the prediction from the linear model with the prediction from the model allowing state dependence. Figure 5 analyzes the response of aggregate consumption to summarize the differences in the predictions from the linear model vis-à-vis the state-dependent model. Having established the difference between the linear and the state-dependent model, Figure 6 focuses on the differential effects of shocks to credit uncertainty in recessions and expansions across all the variables.

16. Our sample extends from 1979Q1 to 2019Q4 and includes periods when the zero lower bound constraint on the nominal interest rate binds. As pointed out in Swanson and Williams (2014) and Gertler and Karadi (2015), the 2-year Treasury yield can be used as an alternative to measure monetary policy when the zero lower bound becomes relevant for the federal funds rate.

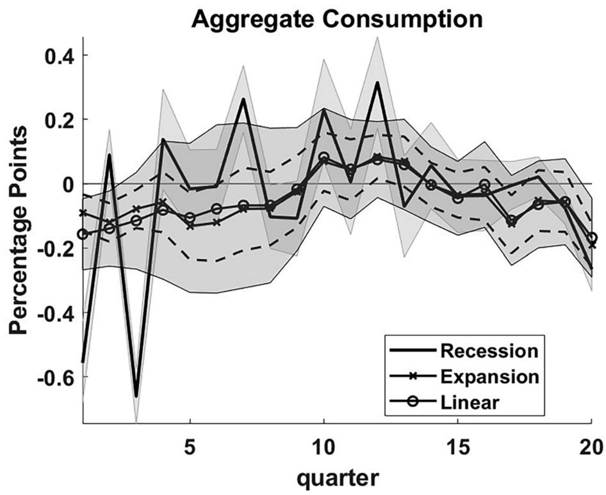

Fig 5. Impact of a One-Standard-Deviation Increase in Credit Uncertainty.

NOTES: Impulse responses are calculated for the quarterly real growth rate of aggregate consumption. The circled black line is the effect of a one-standard-deviation increase in η_t in the linear model (shaded area—68% CI). The black line is the effect of a one-standard-deviation increase in η_t in recessions (shaded area—68% CI). The crossed black line is the effect of a one-standard-deviation increase in η_t in expansions (dashed black lines—68% CI).

Figure 5 shows that a shock to credit uncertainty is contractionary in the linear model. However, decomposing the effects across recessions (black line) and expansions (red line) shows that the effects of credit uncertainty matter largely during downturns where the decline in consumption on impact is about four times larger compared to the linear model and more than six times larger compared to expansions. Section S11 of the online supplement plots the impulse responses for all the macroeconomic variables, comparing the prediction from the linear model along with the responses for recessions and expansions to show that the linear model consistently underestimates the impact of credit uncertainty across all the variables considered in the analysis.

Qualitatively, the results from the linear model align with the empirical regularities characterizing the effects of aggregate uncertainty using alternative measures such as the VIX (Bloom 2009), the JLN index (Jurado, Ludvigson, and Ng 2015), and what the theoretical model predicts. The bigger effect in recessions is similar to those in Chatterjee (2019) and reinforces the importance of state dependence in examining the effects of uncertainty shocks.

We now focus on the asymmetric impact of shocks to credit uncertainty in recessions and expansions in Figure 6. During downturns, macro-economic variables, such as vehicle sales and durable goods consumption, that are relatively more credit-dependent show a sharper decline compared to broader measures of activity, such as aggregate consumption. Nondurable goods consumption also decreases on impact but

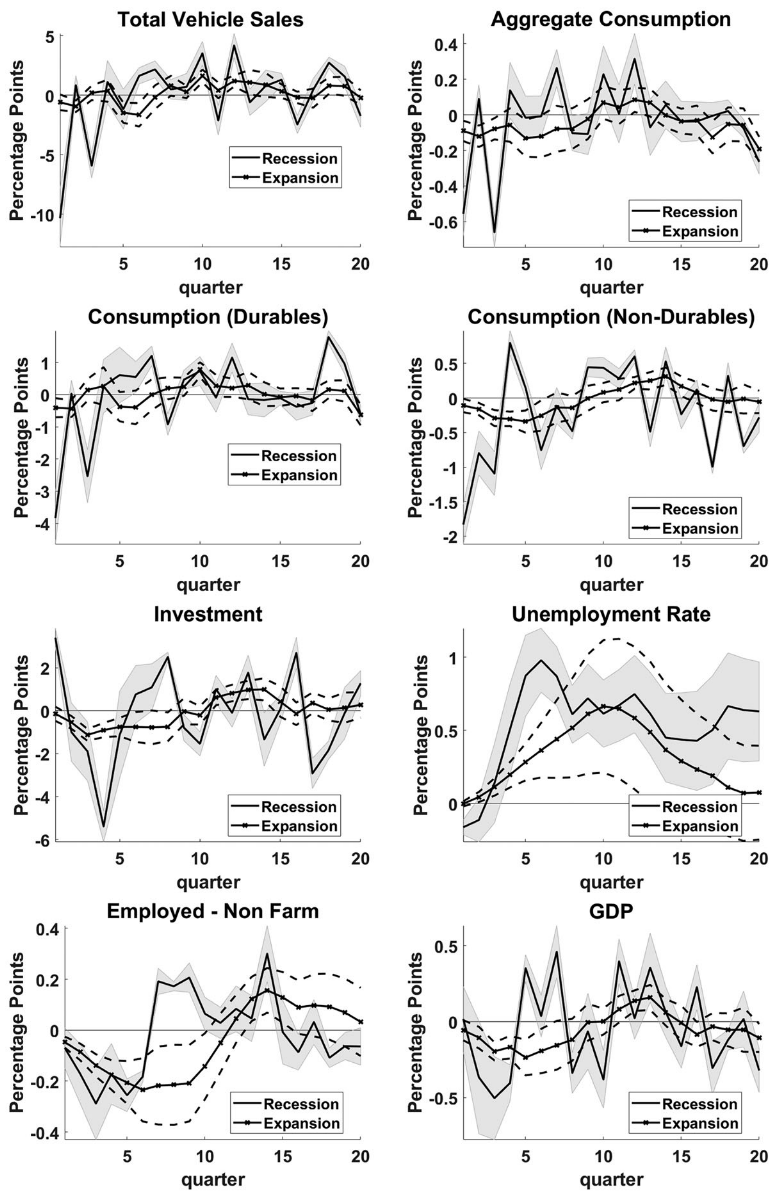

Fig 6. Impact of a One-Standard-Deviation Increase in Credit Uncertainty.

NOTES: Impulse responses are calculated for the quarterly real growth rate of total vehicle sales, durable goods consumption, total consumption, nondurable goods consumption, investment, GDP, the unemployment rate, and total employed in the nonfarm sector. The black line is the effect of a one-standard-deviation increase in η_t in recessions (shaded gray area—68% CI). The crossed black line is the effect of a one-standard-deviation increase in η_t in expansions (dashed black lines—68% CI).

by less than durable goods consumption. Investment, like the credit-sensitive components of consumption, also records a sharper slowdown. The unemployment rate peaks five quarters after the initial shock. In addition to examining the response of the unemployment rate, we also examine the response of total employed and find a significant decline on impact and persistent effects in the quarters following the shock. When the effects across all these measures of activity are combined, we find that GDP declines as well.

In Sections S4.1 and S4.2 of the online supplement, we plot impulse responses of macro-economic variables for an uncertainty shock where the shock is constructed using data on the growth rate in consumer credit and nonfinancial business debt instead of total credit available to the nonfinancial sector as the underlying measure of credit growth rate. Irrespective of variable choice in estimating credit uncertainty, the asymmetry of transmission in recessions and expansions across a broad measure of macro-economic activity is consistently observed.

Figure 6 thus empirically tests the channel presented in Section 3, which examines the impact of credit uncertainty in a flexible-price DSGE model. While the empirical results emphasize the importance of state dependence, which implies asymmetry in how unforeseen changes to credit uncertainty transmit to the economy, it is consistent with the overall patterns we recover from the model.

3.2 Differential Effects of Shocks to Credit Uncertainty

The results in Section 3.1 suggest that the empirical patterns are consistent with what the theory predicts regarding the transmission of unforeseen changes in credit uncertainty. The empirical findings, however, emphasize the importance of state dependence.

We, therefore, conclude our analysis by examining the state-dependent effects of credit uncertainty in the theoretical model. Section 2 solves the model using perturbation methods, assuming that the credit constraint always binds. To evaluate the state-dependent effects of changes in credit uncertainty, we relax this assumption and solve the model using global methods, allowing occasionally binding credit constraints.

To solve the model using global methods, we use the policy function iteration technique described in Richter, Throckmorton, and Walker (2014). We use a linear guess to ease the computational burden when solving the fully nonlinear model, allowing for occasionally binding constraints. To generate this guess, we solve a linear approximation of the model using the OccBin toolkit developed in Guerrieri and Iacoviello (2015) and use that solution as an initial value to the global problem. Initiating the algorithm from this guess also helps in convergence.[17] The nonstochastic steady state of the fully nonlinear model is such that the constraint is just relaxed, with $\mu = 0.0602$.[18] Remaining parameters are calibrated as described in Section 3.3.

17. Section S7 of the online supplement presents details underlying the solution of the model with global methods.

18. In steady state, for the constraint to always bind with

To capture the impact of state dependence in the fully nonlinear model, we construct impulse responses for the same 1% shock to credit uncertainty, however, with different initial conditions underlying the distance to slackness.[19] Unlike the model solved using perturbation methods, differences in initial conditions generate different trajectories for variables in the fully nonlinear model. Specifically, we examine impulse responses using two initial conditions—*State 1,* where the distance to slackness is 0.25% below the value of ζ_t in the nonstochastic steady state where the constraint does not bind, and—*State 2,* where the distance to slackness is 0.75% below the value of ζ_t in the nonstochastic steady state where the constraint does not bind. State 1, in this case, quantifies expansions, and State 2 quantifies recessions.

This approach to characterizing recessions and expansions via differences in distances to slackness underlying the credit constraint is consistent with the evidence in Figure 3, which compares responses from the SLOOS with estimated shocks to credit uncertainty. Responses from the SLOOS suggest a significant spike in credit tightening in recessions. In the context of our model, this can be interpreted as the collateral constraint binding more strongly or, equivalently, the distance to slackness increasing in recessions. This approach to state dependence aligns with Guerrieri and Iacoviello (2017), who find that the collateral constraint on housing wealth became slack in expansions and tightened during the Great Recession. (2013),

Figure 7 presents the differential effects of a 1% shock to credit uncertainty for States 1 (expansions -- distance to slackness of 0.25%) and 2 (recessions -- distance to slackness of 0.75%). Figure 7 demonstrates that the real effects of increases in credit uncertainty are driven substantially by the initial state of the economy—which in this case is the distance to slackness. The greater the distance to slackness on impact, the greater the impact of identical increases in credit uncertainty. Furthermore, a smaller distance to slackness on impact implies a faster recovery with medium-term volatility overshoot in the transmission mechanism and the constraint relaxing faster in the periods following the shock.

$$\mu > 0 \Rightarrow \frac{(1-\alpha)}{\zeta_{ss}}\left(\frac{1}{\beta} - (1-\delta)\right) - \alpha > 0 \Rightarrow \zeta_{ss} < \frac{(1-\alpha)}{\alpha}\left(\frac{1}{\beta} - (1-\delta)\right).$$

$$\text{Plugging, } \alpha = 0.33,\ \beta = 0.99,\ \delta = 0.02 \Rightarrow \zeta_{ss} < 0.0602.$$

19. In Section 2, when the model is solved using perturbation methods, we examine impulse responses for a one-standard-deviation increase in credit uncertainty, which in percent terms implies about a 26% increase in uncertainty. The size of one-standard-deviation shock in the model solved using a third-order approximation is bigger relative to what we present for the fully global model; however, it is consistent with other papers that examine the impact of uncertainty (Basu and Bundick (2017) estimate that a 100% increase in uncertainty underlying a demand shock is needed to generate a 15% increase in the model implied stock market volatility). We consider a smaller shock size for examining impulse responses in the fully nonlinear solved using global methods. This choice is driven by computational feasibility. We need to consider bigger and denser grids to accurately examine the impact of bigger shocks on credit uncertainty. To keep the computational time reasonable, we demonstrate the impact of credit uncertainty with a smaller shock. Note that the qualitative features guiding the transmission of credit uncertainty in the model solved using global methods are comparable to what we present with the model solved using perturbation methods.

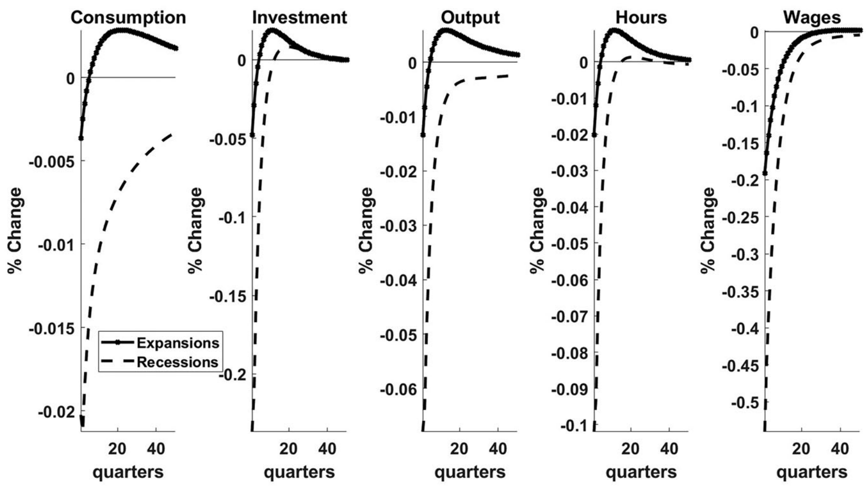

Fig 7. Examining the Impulse Responses of Consumption, Investment, GDP, Hours, and Wages for the Same Uncertainty Shock But Different States—Expansions and Recessions.

NOTES: The crossed line plots the responses when the constraint is 0.25% (expansions) away from slackness, and the dashed line plots responses when the constraint is 0.75% (recessions) away from slackness.

As the distance to slackness varies, the transmission mechanism describing the comovement in variables in the model remains unchanged; however, the intensity of responses gets smaller as we move closer to the constraint becoming slack. To highlight this feature, we examine the response on impact for GDP (output) in Figure 7. In recessions, when the constraint is farther away from slackness, the decline in GDP is about six times more compared to expansions when the constraint is closer to slackness. We observe a similar pattern in the other variables as well.

Qualitatively, the results are consistent with what we obtain from the model solved using perturbation methods in Section 3.4; however, by solving the fully nonlinear model, with occasionally binding constraints, we generate state-dependent effects of changes in credit uncertainty consistent with the empirical evidence in Figure 6. Figure 7 thus also demonstrates that the results are robust to the choice of solution technique for the DSGE model.

4. CONCLUSION

This paper presents new stylized facts characterizing the time variation in the volatility of the credit growth rate. We interpret this stochastic volatility in credit growth as credit uncertainty. Motivated by the evidence of SV underlying credit growth rate in the data, we introduce a role for credit uncertainty via collateral constraints in an otherwise standard RBC model.When the collateral constraint binds, unforeseen changes in credit uncertainty transmit by triggering a precautionary response that interacts with the collateral constraint to generate a sizable and simultaneous decline in output, consumption, investment, real wages, and hours.

Furthermore, we empirically test the models' predictions and show that unforeseen changes in credit uncertainty propagate via contraction in a broad measure of real activity—these effects are asymmetric—and matter mostly during downturns. We solve our proposed model using global methods to show that by varying the initial condition describing the distance to slackness underlying the collateral constraint, the framework can generate the asymmetry in the responses of macro variables in recessions and expansions consistent with the empirical patterns. This interaction between credit uncertainty, precautionary motives, and credit constraints is a novel feature that delivers comovement in macro-economic variables, which previous work on uncertainty shocks without credit constraints has been unable to produce in a flexible-price environment.

LITERATURE CITED

Alessandri, Piergiorgio, and Haroon Mumtaz. (2019) "Financial Regimes and Uncertainty Shocks." *Journal of Monetary Economics*, 101, 31–46.

Alfaro, Ivan, Nicholas Bloom, and Xiaoji Lin. (2024) "The Finance Uncertainty Multiplier." *Journal of Political Economy*, 132, 577–615.

Andreasen, Martin M. (2012) "On the Effects of Rare Disasters and Uncertainty Shocks for Risk Premia in Non-Linear DSGE Models." *Review of Economic Dynamics*, 15, 295–316.

Andreasen, Martin M., Jesús Fernández-Villaverde, and Juan F. Rubio-Ramírez. (2018) "The Pruned State-Space System for Non-Linear DSGE Models: Theory and Empirical Applications." *Review of Economic Studies*, 85, 1–49.

Arellano, Cristina, Yan Bai, and Patrick J. Kehoe. (2019) "Financial Frictions and Fluctuations in Volatility." *Journal of Political Economy*, 127, 2049–102.

Baker, Scott R., Nicholas Bloom, and Steven J. Davis. (2016) "Measuring Economic Policy Uncertainty." *Quarterly Journal of Economics*, 131, 1593–636.

Basu, Susanto, and Brent Bundick. (2017) "Uncertainty Shocks in a Model of Effective Demand." *Econometrica*, 85, 937–58.

Berger, David, Ian Dew-Becker, and Stefano Giglio. (2020) "Uncertainty Shocks as Second-Moment News Shocks." *Review of Economic Studies*, 87, 40–76.

Bloom, Nicholas. (2009) "The Impact of Uncertainty Shocks." *Econometrica*, 77, 623–85.

Born, Bejamin, and Johannes Pfeifer. (2021) "Uncertainty-Driven Business Cycles: Assessing the Markup Channel." *Quantitative Economics*, 12, 587–623.

Brave, Scott, and Andrew Butters. (2012) "Diagnosing the Financial System: Financial Conditions and Financial Stress." *International Journal of Central Banking*, 8, 191–239.

Caggiano, Giovanni, Efrem Castelnuovo, Silvia Delrio, and Richard Kima. (2020) "Financial Uncertainty and Real Activity: The Good, the Bad, and the Ugly." CAMA Working Paper, 67.

Caldara, Dario, Cristina Fuentes-Albero, Simon Gilchrist, and Egon Zakrajšek. (2016) “The Macroeconomic Impact of Financial and Uncertainty Shocks.” *European Economic Review*, 88, 185–207.

Carlstrom, Charles T., and Timothy S. Fuerst. (1997) “Agency Costs, Net Worth, and Business Fluctuations: A Computable General Equilibrium Analysis.” *American Economic Review*, 87, 893–910.

Chatterjee, Pratiti. (2019) “Asymmetric Impact of Uncertainty in Recessions: Are Emerging Countries More Vulnerable?” *Studies in Nonlinear Dynamics & Econometrics*, 23, 1–27.

Christiano, Lawrence J., Roberto Motto, and Massimo Rostagno. (2014) “Risk Shocks.” *American Economic Review*, 104, 27–65.

Deligiannidis, George, Arnaud Doucet, and Michel K. Pitt. (2018) “The Correlated Pseudo Marginal Method.” *Journal of Royal Statistical Society Series B*, 80, 839–70.

Fernández-Villaverde, Jesús, and Pablo Guerrón-Quintana. (2020) “Uncertainty Shocks and Business Cycle Research.” *Review of Economic Dynamics*, 37, S118–46.

Fernández-Villaverde, Jesús, Pablo Guerrón-Quintana, Juan F. Rubio-Ramírez, and Martin Uribe. (2011) “Risk Matters: The Real Effects of Volatility Shocks.” *American Economic Review*, 101, 2530–61.

Fernández-Villaverde, Jesús, Pablo Guerrón-Quintana, Keith Kuester, and Juan F. Rubio-Ramírez. (2015) “Fiscal Volatility Shocks and Economic Activity.” *American Economic Review*, 105, 3352–84.

Fisher, Richard W. (2013) “Uncertainty Matters.” Remarks before the Causes & Macroeconomic Consequences of Uncertainty Conference.

Garthwaite, Paul H., Yanan Fan, and Scott A. Sisson. (2016) “Adaptive Optimal Scaling of Metropolis–Hastings Algorithms using the Robbins–Monro Process.” *Communications in Statistics - Theory and Methods*, 45, 5098–111.

Gertler, Mark, and Peter Karadi. (2015) “Monetary Policy Surprises, Credit Costs, and Economic Activity.” *American Economic Journal: Macroeconomics*, 7, 44–76.

Gopinath, Gita. (2020) *World Economic Outlook, April 2020: The Great Lockdown*. International Monetary Fund, Washington.

Guerrieri, Luca, and Matteo Iacoviello. (2015) “Collateral Constraints and Macroeconomic Asymmetries.” *Journal of Monetary Economics*, 70, 22–38.

Guerrieri, Luca, and Matteo Iacoviello. (2017) “Collateral Constraints and Macroeconomic Asymmetries.” *Journal of Monetary Economics*, 90, 28–49.

Jensen, Henrik, Ivan Petrella, Søren Hove Ravn, and Emiliano Santoro. (2020) “Leverage and Deepening Business-Cycle Skewness.” *American Economic Journal: Macroeconomics*, 12, 245–81.

Jermann, Urban, and Vincenzo Quadrini. (2012) “Macroeconomic Effects of Financial Shocks.” *American Economic Review*, 102, 238–71.

Òscar, Jordà,<fnm>. (2005) “Estimation and Inference of Impulse Responses by Local Projections.” *American Economic Review*, 95, 161–82.

Jordà, Òscar, Mortiz Schularick, and Alan M. Taylor. (2013) “When Credit Bites Back.” *Journal of Money, Credit and Banking*, 45, 3–28.

Jurado, Kyle, Sydney C. Ludvigson, and Serena Ng. (2015) “Measuring Uncertainty.” *American Economic Review*, 105, 1177–216.

Kim, Sangjoon, Neil Shephard, and Siddhartha Chib. (1998) "Stochastic Volatility: Likelihood Inference and Comparison with ARCH Models." *Review of Economic Studies*, 65, 361–93.

Kiyotaki, Nobuhiro, and John Moore. (1997) "Credit Cycles." *Journal of Political Economy*, 105, 211–48.

Lagarde, Christine. (2015) "IMF Survey: Lagarde Calls for Policy Upgrade to Combat Uncertain Global Outlook." International Monetary Fund, Washington.

Leduc, Sylvain, and Zheng Liu. (2016) "Uncertainty Shocks are Aggregate Demand Shocks." *Journal of Monetary Economics*, 82, 20–35.

Liu, Zheng, Pengfei Wang, and Tao Zha. (2013) "Land-Price Dynamics and Macroeconomic Fluctuations." *Econometrica*, 81, 31–46.

Ludvigson, Sydney C., Sai Ma, and Serena Ng. (2021) "Uncertainty and Business Cycles: Exogenous Impulse or Endogenous Response?" *American Economic Journal: Macroeconomics*, 13, 369–410.

Mian, Atif, and Amir Sufi. (2018) "Finance and Business Cycles: The Credit- Driven Household Demand Channel." *Journal of Economic Perspectives*, 32, 31–58.

Newey, Whitney K., and Kenneth D. West. (1987) "A Simple, Positive Semidefinite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix." *Econometrica*, 55, 703–8.

Ramey, Valerie A., and Sarah Zubairy. (2018) "Government Spending Multipliers in Good Times and in Bad: Evidence from US Historical Data." *Journal of Political Economy*, 126, 850–901.

Richter, Alexander W., Nathaniel Throckmorton, and Todd Walker. (2014) "Accuracy, Speed and Robustness of Policy Function Iteration." *Computational Economics*, 44, 445–76.

Schmitt-Grohé, Stephanie, and Martin Uribe. (2004) "Solving Dynamic General Equilibrium Models Using a Second-Order Approximation to the Policy Function." *Journal of Economic Dynamics & Control*, 28, 755–75.

Sims, Eric. (2017) "The Basics of Financial Constraints." Mimeo.

Swanson, Eric T., and John C. Williams. (2014) "Measuring the Effect of the Zero Lower Bound on Medium- and Longer-Term Interest Rates." *American Economic Review*, 104, 3154–85.

SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

Table S1: Posterior mean estimates (with 95% credible intervals in brackets) of the standard SV model without leverage and the SV model with leverage

Table S2: Robustness checks.

Figure S1: Impact of changes in η_t on real activity.

Figure S2: Impact of changes in η_t on real activity.

Figure S3: Examining the impact of an increase in credit uncertainty using the framework proposed in Jermann and Quadrini (2012).

Figure S4: Plotting changes in credit uncertainty and the occasionally binding constraint in the fully nonlinear model.

Figure S5: Shock to credit-uncertainty (ht) when constraint binds in the initial period

Figure S6: Comparing responses between the fully global solution allowing for occasionally binding constraints and the solution obtained using a third order approximation of equilibrium conditions with the constraint always binding for the same uncertainty shock.

Figure S7: Generating trajectories of the model variables with a recession in the initial period.

Figure S8: Examining the evolution of ζ_t and the impulse response of labor-wedge (μ_t) for the same uncertainty shock but different distances to slackness.

Figure S9: Examining the impulse responses of consumption, investment, GDP, hours, and wages for the same uncertainty shock but different distances to slackness.

Figure S10: Impact of changes in η_t on real activity.